

# Report on DiagMC Lab

## Student group report

### Student and group information

|                              |                                                                                                                        |
|------------------------------|------------------------------------------------------------------------------------------------------------------------|
| <b>Name:</b>                 | <i>Student name</i>                                                                                                    |
| <b>Family name:</b>          | <i>Student family name</i>                                                                                             |
| <b>Matriculation number:</b> | <i>Student matriculation number</i>                                                                                    |
| <b>Group number:</b>         | <i>Group number</i>                                                                                                    |
| <b>Group members:</b>        | <i>Name, family name, and matriculation number of each member of the group. Groups should contain 2 or 3 students.</i> |
| <b>Date:</b>                 | <i>Date of submission</i>                                                                                              |

## Contents

|          |                                                 |           |
|----------|-------------------------------------------------|-----------|
| <b>1</b> | <b>Exercise 1: <i>Title of the activity</i></b> | <b>2</b>  |
| 1.1      | Introduction . . . . .                          | 2         |
| 1.2      | Computational Method . . . . .                  | 2         |
| 1.3      | Results and Discussion . . . . .                | 3         |
| <b>2</b> | <b>Exercise 2: <i>Title of the activity</i></b> | <b>4</b>  |
| 2.1      | Introduction . . . . .                          | 4         |
| 2.2      | Computational Method . . . . .                  | 4         |
| 2.3      | Results and Discussion . . . . .                | 5         |
| <b>3</b> | <b>Exercise 3: <i>Title of the activity</i></b> | <b>6</b>  |
| 3.1      | Introduction . . . . .                          | 6         |
| 3.2      | Computational Method . . . . .                  | 6         |
| 3.3      | Results and Discussion . . . . .                | 7         |
| <b>4</b> | <b>Exercise 4: <i>Title of the activity</i></b> | <b>8</b>  |
| 4.1      | Introduction . . . . .                          | 8         |
| 4.2      | Computational Method . . . . .                  | 8         |
| 4.3      | Results and Discussion . . . . .                | 9         |
| <b>5</b> | <b>Exercise 5: <i>Title of the activity</i></b> | <b>10</b> |
| 5.1      | Introduction . . . . .                          | 10        |
| 5.2      | Computational Method . . . . .                  | 10        |

|                                      |           |
|--------------------------------------|-----------|
| 5.3 Results and Discussion . . . . . | 11        |
| <b>Optional: References</b>          | <b>12</b> |

# 1 Exercise 1: *Title of the activity*

## 1.1 Introduction

*Introduce the goal of Exercise 1. Briefly explain the physical, mathematical, or computational problem addressed in this activity. State what quantity, property, or behavior is expected to be calculated, simulated, or analyzed.*

Suggested points to include:

- the scientific or computational objective of the exercise;
- the system, model, or algorithm considered;
- the main question that the calculation or simulation should answer;
- the expected outcome or observable, when known.

## 1.2 Computational Method

*Describe the computational protocol used in Exercise 1. Introduce the code or software used, explain what the code is designed to do, and summarize the main computational steps followed during the work.*

The description may include:

- name and brief description of the code/software used;
- input parameters and relevant settings;
- initial configuration or initial conditions;
- sequence of calculations or simulations performed;
- convergence criteria, approximations, or numerical parameters;
- post-processing or data-analysis steps.

Important computational parameters can be summarized in a table such as Table 1.

Table 1: Main computational parameters for Exercise 1. Replace the entries with those used in the exercise.

| Parameter          | Value        | Comment              |
|--------------------|--------------|----------------------|
| <i>Parameter 1</i> | <i>Value</i> | <i>Short comment</i> |
| <i>Parameter 2</i> | <i>Value</i> | <i>Short comment</i> |
| <i>Parameter 3</i> | <i>Value</i> | <i>Short comment</i> |

### 1.3 Results and Discussion

*Present the results obtained in Exercise 1. Include the necessary tables, figures, numerical values, plots, and comments. The discussion should explain the meaning of the results and connect them to the goal stated in the Introduction.*

When presenting figures, use clear captions and refer to each figure in the text. An example figure placeholder is shown in Fig. 1.

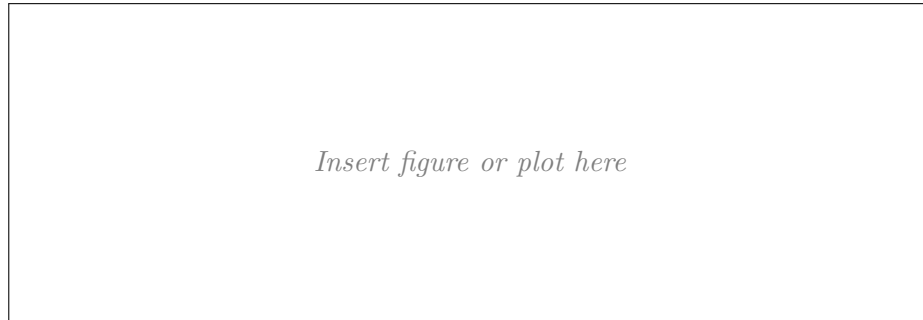


Figure 1: Example figure caption for Exercise 1. Replace this placeholder with a concise description of the figure and the main information it contains.

The discussion should address the following points:

- What are the main numerical or qualitative results?
- Are the results consistent with expectations, theory, or reference data?
- Were the goals of this exercise fully fulfilled?
- Were there any difficulties, critical issues, incomplete parts, or open aspects?

*Write here the final discussion of the results for Exercise 1. Include comments on accuracy, limitations, possible sources of error, and any aspect of the exercise that was not completed or requires further investigation.*

## 2 Exercise 2: *Title of the activity*

### 2.1 Introduction

*Introduce the goal of Exercise 2. Briefly explain the physical, mathematical, or computational problem addressed in this activity. State what quantity, property, or behavior is expected to be calculated, simulated, or analyzed.*

Suggested points to include:

- the scientific or computational objective of the exercise;
- the system, model, or algorithm considered;
- the main question that the calculation or simulation should answer;
- the expected outcome or observable, when known.

### 2.2 Computational Method

*Describe the computational protocol used in Exercise 2. Introduce the code or software used, explain what the code is designed to do, and summarize the main computational steps followed during the work.*

The description may include:

- name and brief description of the code/software used;
- input parameters and relevant settings;
- initial configuration or initial conditions;
- sequence of calculations or simulations performed;
- convergence criteria, approximations, or numerical parameters;
- post-processing or data-analysis steps.

Important computational parameters can be summarized in a table such as Table 2.

Table 2: Main computational parameters for Exercise 2. Replace the entries with those used in the exercise.

| Parameter          | Value        | Comment              |
|--------------------|--------------|----------------------|
| <i>Parameter 1</i> | <i>Value</i> | <i>Short comment</i> |
| <i>Parameter 2</i> | <i>Value</i> | <i>Short comment</i> |
| <i>Parameter 3</i> | <i>Value</i> | <i>Short comment</i> |

## 2.3 Results and Discussion

*Present the results obtained in Exercise 2. Include the necessary tables, figures, numerical values, plots, and comments. The discussion should explain the meaning of the results and connect them to the goal stated in the Introduction.*

When presenting figures, use clear captions and refer to each figure in the text. An example figure placeholder is shown in Fig. 2.

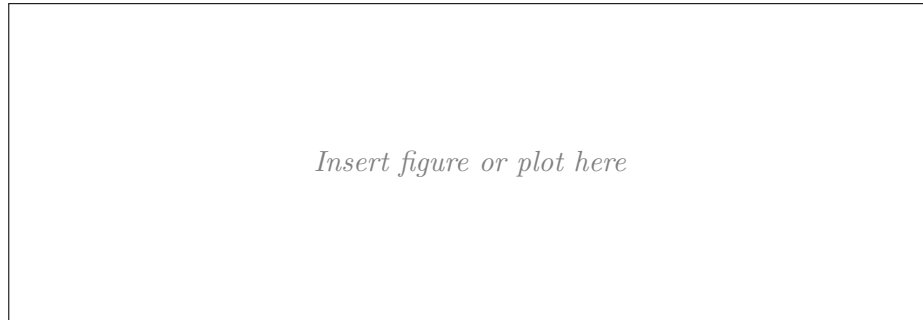


Figure 2: Example figure caption for Exercise 2. Replace this placeholder with a concise description of the figure and the main information it contains.

The discussion should address the following points:

- What are the main numerical or qualitative results?
- Are the results consistent with expectations, theory, or reference data?
- Were the goals of this exercise fully fulfilled?
- Were there any difficulties, critical issues, incomplete parts, or open aspects?

*Write here the final discussion of the results for Exercise 2. Include comments on accuracy, limitations, possible sources of error, and any aspect of the exercise that was not completed or requires further investigation.*

### 3 Exercise 3: *Title of the activity*

#### 3.1 Introduction

*Introduce the goal of Exercise 3. Briefly explain the physical, mathematical, or computational problem addressed in this activity. State what quantity, property, or behavior is expected to be calculated, simulated, or analyzed.*

Suggested points to include:

- the scientific or computational objective of the exercise;
- the system, model, or algorithm considered;
- the main question that the calculation or simulation should answer;
- the expected outcome or observable, when known.

#### 3.2 Computational Method

*Describe the computational protocol used in Exercise 3. Introduce the code or software used, explain what the code is designed to do, and summarize the main computational steps followed during the work.*

The description may include:

- name and brief description of the code/software used;
- input parameters and relevant settings;
- initial configuration or initial conditions;
- sequence of calculations or simulations performed;
- convergence criteria, approximations, or numerical parameters;
- post-processing or data-analysis steps.

Important computational parameters can be summarized in a table such as Table 3.

Table 3: Main computational parameters for Exercise 3. Replace the entries with those used in the exercise.

| Parameter          | Value        | Comment              |
|--------------------|--------------|----------------------|
| <i>Parameter 1</i> | <i>Value</i> | <i>Short comment</i> |
| <i>Parameter 2</i> | <i>Value</i> | <i>Short comment</i> |
| <i>Parameter 3</i> | <i>Value</i> | <i>Short comment</i> |

### 3.3 Results and Discussion

*Present the results obtained in Exercise 3. Include the necessary tables, figures, numerical values, plots, and comments. The discussion should explain the meaning of the results and connect them to the goal stated in the Introduction.*

When presenting figures, use clear captions and refer to each figure in the text. An example figure placeholder is shown in Fig. 3.

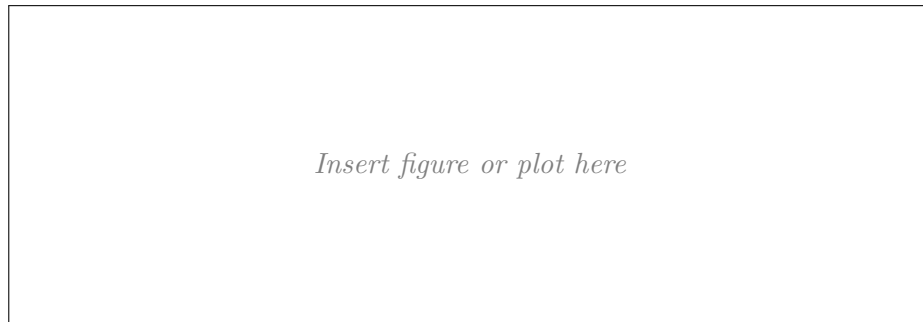


Figure 3: Example figure caption for Exercise 3. Replace this placeholder with a concise description of the figure and the main information it contains.

The discussion should address the following points:

- What are the main numerical or qualitative results?
- Are the results consistent with expectations, theory, or reference data?
- Were the goals of this exercise fully fulfilled?
- Were there any difficulties, critical issues, incomplete parts, or open aspects?

*Write here the final discussion of the results for Exercise 3. Include comments on accuracy, limitations, possible sources of error, and any aspect of the exercise that was not completed or requires further investigation.*



## 4 Exercise 4: *Title of the activity*

### 4.1 Introduction

*Introduce the goal of Exercise 4. Briefly explain the physical, mathematical, or computational problem addressed in this activity. State what quantity, property, or behavior is expected to be calculated, simulated, or analyzed.*

Suggested points to include:

- the scientific or computational objective of the exercise;
- the system, model, or algorithm considered;
- the main question that the calculation or simulation should answer;
- the expected outcome or observable, when known.

### 4.2 Computational Method

*Describe the computational protocol used in Exercise 4. Introduce the code or software used, explain what the code is designed to do, and summarize the main computational steps followed during the work.*

The description may include:

- name and brief description of the code/software used;
- input parameters and relevant settings;
- initial configuration or initial conditions;
- sequence of calculations or simulations performed;
- convergence criteria, approximations, or numerical parameters;
- post-processing or data-analysis steps.

Important computational parameters can be summarized in a table such as Table 4.

Table 4: Main computational parameters for Exercise 4. Replace the entries with those used in the exercise.

| Parameter          | Value        | Comment              |
|--------------------|--------------|----------------------|
| <i>Parameter 1</i> | <i>Value</i> | <i>Short comment</i> |
| <i>Parameter 2</i> | <i>Value</i> | <i>Short comment</i> |
| <i>Parameter 3</i> | <i>Value</i> | <i>Short comment</i> |

### 4.3 Results and Discussion

*Present the results obtained in Exercise 4. Include the necessary tables, figures, numerical values, plots, and comments. The discussion should explain the meaning of the results and connect them to the goal stated in the Introduction.*

When presenting figures, use clear captions and refer to each figure in the text. An example figure placeholder is shown in Fig. 4.

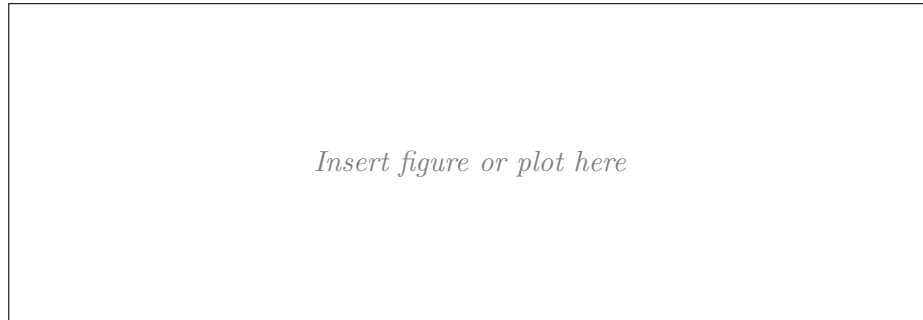


Figure 4: Example figure caption for Exercise 4. Replace this placeholder with a concise description of the figure and the main information it contains.

The discussion should address the following points:

- What are the main numerical or qualitative results?
- Are the results consistent with expectations, theory, or reference data?
- Were the goals of this exercise fully fulfilled?
- Were there any difficulties, critical issues, incomplete parts, or open aspects?

*Write here the final discussion of the results for Exercise 4. Include comments on accuracy, limitations, possible sources of error, and any aspect of the exercise that was not completed or requires further investigation.*

## 5 Exercise 5: *Title of the activity*

### 5.1 Introduction

*Introduce the goal of Exercise 5. Briefly explain the physical, mathematical, or computational problem addressed in this activity. State what quantity, property, or behavior is expected to be calculated, simulated, or analyzed.*

Suggested points to include:

- the scientific or computational objective of the exercise;
- the system, model, or algorithm considered;
- the main question that the calculation or simulation should answer;
- the expected outcome or observable, when known.

### 5.2 Computational Method

*Describe the computational protocol used in Exercise 5. Introduce the code or software used, explain what the code is designed to do, and summarize the main computational steps followed during the work.*

The description may include:

- name and brief description of the code/software used;
- input parameters and relevant settings;
- initial configuration or initial conditions;
- sequence of calculations or simulations performed;
- convergence criteria, approximations, or numerical parameters;
- post-processing or data-analysis steps.

Important computational parameters can be summarized in a table such as Table 5.

Table 5: Main computational parameters for Exercise 5. Replace the entries with those used in the exercise.

| Parameter          | Value        | Comment              |
|--------------------|--------------|----------------------|
| <i>Parameter 1</i> | <i>Value</i> | <i>Short comment</i> |
| <i>Parameter 2</i> | <i>Value</i> | <i>Short comment</i> |
| <i>Parameter 3</i> | <i>Value</i> | <i>Short comment</i> |

### 5.3 Results and Discussion

*Present the results obtained in Exercise 5. Include the necessary tables, figures, numerical values, plots, and comments. The discussion should explain the meaning of the results and connect them to the goal stated in the Introduction.*

When presenting figures, use clear captions and refer to each figure in the text. An example figure placeholder is shown in Fig. 5.

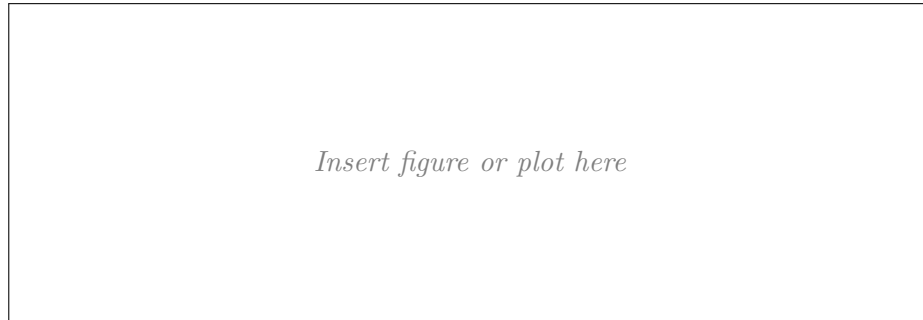


Figure 5: Example figure caption for Exercise 5. Replace this placeholder with a concise description of the figure and the main information it contains.

The discussion should address the following points:

- What are the main numerical or qualitative results?
- Are the results consistent with expectations, theory, or reference data?
- Were the goals of this exercise fully fulfilled?
- Were there any difficulties, critical issues, incomplete parts, or open aspects?

*Write here the final discussion of the results for Exercise 5. Include comments on accuracy, limitations, possible sources of error, and any aspect of the exercise that was not completed or requires further investigation.*

## Optional: References

*Include references if external sources, manuals, papers, lecture notes, or online documentation were used.*